Train-Test Split

Splitting the dataset into training and testing sets is an essential step in machine learning. The training dataset is used to train the model so that it can learn patterns and relationships between the input features and the target variable. During this phase, the model adjusts its internal parameters to improve prediction performance.

The testing dataset is used to evaluate the trained model on unseen data. This helps determine how well the model generalizes to new data rather than memorizing the training examples. Evaluation metrics such as Accuracy, Precision, Recall, F1-Score, MAE, MSE, RMSE, or R^2 Score can then be calculated depending on the type of machine learning problem.

Code:

```
#Train and test split
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.1, random_state=42)
```

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